Katie Spoon

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Research Interests

Computational social science — big data, administrative records, surveys, networks, text analysis
Social inequalities — in education, careers, health, and their policy implications
Science of science — faculty retention, workforce development, institutional prestige
Diversity, equity and inclusion (DEI) — marginalized experiences in science, backlash to DEI, policy
Deep learning — AI for social good, computer vision, natural language processing

Education

Ph.D. in Computer Science, University of Colorado Boulder 2020 – 2025
Advisors: Aaron Clauset & Dan Larremore
Dissertation: *Quantifying Systemic Inequities in the Training and Retention of Professors*

M.A. in Education Policy, University of Colorado Boulder 2022 – 2024
Advisor: Kevin Welner
Thesis: *Quantifying Inequitable Education Pathways to STEM Careers*

M.S. in Computer Science, Indiana University Bloomington 2018 – 2019
Advisors: David Crandall & Katie Siek
Thesis: *Detecting Dyslexia in Handwriting Using Neural Networks*

B.S. in Computer Science with highest honors, Minor: Math, Indiana University Bloomington 2015 – 2019

Selected Employment

Data Science Fellow and **New Map of Life Fellow** 2025 –
Stanford University, *Stanford Data Science* and *Stanford Center on Longevity* (Stanford, CA)

Data Science Fellow via Coding it Forward; 80 fellows selected from 4,500+ applications June – Aug 2024
U.S. Census Bureau, *Enhancing Health Data Group* (Remote)

Graduate Research Assistant 2024 – 2025
National Center for Women & Information Technology (Boulder, CO)

Research Engineer June 2019 – Aug 2020
IBM Research, *Artificial Intelligence Hardware Group* (San Jose, CA)

Honors & Awards

Achievement Reward for College Scientists, ARCS Foundation 2024 – 2025
$7,500 award of "unrestricted funding to create new knowledge & innovative technology"

Rising Star in Data Science, UC San Diego, UChicago, and Stanford 2024
"Celebrating and fast-tracking the careers of exceptional data scientists"

NSF Graduate Research Fellowship 2021 – 2024
$37,000/year and tuition for three years of graduate school

National Center for Women in Technology Collegiate Award 2019
$10,000 award "recognizing technical contributions to projects with high innovation & impact"

Provost's Award for Outstanding Undergraduate Research and Creative Activity, Indiana Univ. 2019
Mathematics & Natural Sciences winner, one of five categories total

Graduate Teaching Assistant of the Year, Indiana Univ. 2019

Emerging Research Scholar, Center of Excellence for Women & Technology, Indiana Univ. 2015 – 2016

Publication & Presentation Awards

Annual Outstanding Research Paper, Univ. Colorado 2024
Bell Family Endowed Computer Science Award, "recognizing outstanding research", Univ. Colorado 2024
Research Expo Winner, Univ. Colorado 2024
Publication Recognition Award, Univ. Colorado 2023
Best Poster Award, International Conference on Machine Learning 2019
Highlighted Proposal, Global Challenges Workshop, Computer Vision and Pattern Recognition 2019
Exemplary paper in social and interdisciplinary sciences, *Science Advances* 2023

Publications

* contributed equally † listed alphabetically ° mentee coauthor

In Preparation

1. Backlash to diversity, equity and inclusion efforts among faculty 2024
   K. Spoon, °J. Mendy, L. Ekpe, and L. A. Rivera. Draft available upon request.

2. The elite undergraduate backgrounds of U.S. professors 2024
   *__K. Spoon__, *E. Lee, A. Clauset, D. B. Larremore. Draft available upon request.

Peer-Reviewed Journal Articles

3. Devaluation underlies gendered U.S. faculty departures across disciplines and institutions 2026
   K. Spoon, °J. Mendy, °M. Martinez, M. Galesic, D. B. Larremore, A. Clauset, L. A. Rivera. *PNAS Nexus*.

4. Book bans in political context: Evidence from U.S. public schools 2024
   † *M. S. O. Goncalves, *I. Langrock, *J. LaViolette, *__K. Spoon__. *PNAS Nexus*.

5. Gender and retention patterns among U.S. faculty 2023
   K. Spoon, N. LaBerge, K. H. Wapman, S. Zhang, A. C. Morgan, M. Galesic, B. K. Fosdick, D. B. Larremore, and A. Clauset. *Science Advances*.
   Annual Outstanding Research Paper, 2024 · Bell Family Endowed Computer Science Award, 2024 · Publication Recognition Award, 2023

6. Towards software-equivalent accuracy on transformer-based deep neural networks with analog memory devices 2021
   K. Spoon, H. Tsai, A. Chen, M. J. Rasch, S. Ambrogio, C. Mackin, A. Fasoli, A. Friz, P. Narayanan, M. Stanisavljevic, and G. W. Burr. *Frontiers in Computational Neuroscience*.

7. Noise-resilient DNN: Tolerating noise in PCM-based AI accelerators via noise-aware training 2021
   S. Kariyappa, H. Tsai, **K. Spoon**, S. Ambrogio, P. Narayanan, C. Mackin, A. Chen, M. Quereshi, and G. W. Burr. *IEEE Transactions on Electron Devices*.

Peer-Reviewed Conference Papers

8. NSF EHR Core Research: Exploring faculty accountability within departmental change efforts 2025
   W. M. DuBow, H. Thiry, **K. Spoon**. *ASEE Annual Conference & Exposition*.

9. Mushroom-type phase change memory with projection liner: An array-level demonstration of conductance drift and noise mitigation 2021
   R. L. Bruce, et al. [including **K. Spoon**]. *IEEE International Reliability Physics Symposium* (IRPS).

10. Fully on-chip MAC at 14nm enabled by accurate row-wise programming of PCM-based weights and parallel vector-transport in duration-format 2021
    P. Narayanan, et al. [including **K. Spoon**]. *Symposium on VLSI Technology*.

11. Neuromorphic computing with phase change, device reliability, and variability challenges 2020
    C. Mackin, et al. [including **K. Spoon**]. *IEEE International Reliability Physics Symposium* (IRPS).

12. Reducing the impact of phase-change memory conductance drift on the inference of large-scale hardware neural networks 2019
S. Ambrogio, M. Gallot, **K. Spoon**, H. Tsai, C. Mackin, M. Wesson, S. Kariyappa, P. Narayanan, C. C. Liu, A. Kumar, A. Chen, and G. W. Burr. *65th IEEE International Electron Devices Meeting (IEDM)*.
Ranked 2nd / 98 papers

Peer-Reviewed Workshop Papers

13. Accelerating deep neural networks with analog memory devices 2020
K. Spoon, S. Ambrogio, P. Narayanan, H. Tsai, C. Mackin, A. Chen, A. Fasoli, A. Friz, and G. W. Burr. *International Memory Workshop*.

14. Can we (and should we) use AI to detect dyslexia in children's handwriting? 2019
K. Spoon, D. Crandall, K. Siek, and M. Fillmore. *AI for Social Good Workshop, NeurIPS*.

15. Towards detecting dyslexia in children's handwriting using neural networks 2019
K. Spoon, D. Crandall, and K. Siek. *AI for Social Good Workshop, International Conference on Machine Learning (ICML)*.
Best poster award

Book Chapters

16. Accelerating deep neural networks with analog memory devices 2022
K. Spoon, S. Ambrogio, P. Narayanan, H. Tsai, C. Mackin, A. Chen, A. Fasoli, A. Friz and G. W. Burr. In *Machine Learning & Non-Volatile Memories*. Ed. C. Zambelli, Springer.

Essays and Perspectives

17. Most US book bans target children's literature featuring diverse characters and authors of color 2024
K. Spoon. *The Conversation*.

18. Gendered devaluation and retention among U.S. faculty 2024
K. Spoon and A. Clauset. *Committee on the Status of Women in the Economics Profession (CSWEP) News* 2, 7–10.

Dissertation

19. Quantifying systemic inequities in the training and retention of professors 2025
K. Spoon. *University of Colorado Boulder*.

Grants

Quantifying the origins and impacts of book bans in U.S. schools, $1,500 2022
PI, with Isabelle Langrock (co-PI), Jack LaViolette (co-PI) and Marcelo S. O. Goncalves (co-PI)
Russell Sage Foundation & Social Science Research Council

Travel Funding

Conference Support Fellowship, CU Boulder, $2,000 2024
Graduate Student Travel Grant, CU Boulder, $450 2024
Graduate Student Travel Grant, CU Boulder, $450 2023
Summer Institute in Computational Social Science, full travel funding 2022
CVPR Global Challenges Workshop, full travel funding 2019
ICML AI for Social Good Workshop, travel scholarship & registration fee waiver 2019
NCWIT annual conference, full travel funding 2019
Grace Hopper Celebration of Women in Computing, full travel funding 2016

Presentations

Health across the life course

IAGG World Congress of Gerontology and Geriatrics, Amsterdam, Talk	2026
Society for Longitudinal and Life Course Studies, Brussels, Talk	2026
Sociological Science Conference, Palo Alto, Talk	2026
Data-Intensive Research Conference, Minneapolis, Talk	2026

Gendered retention patterns in academia

SCL Futures Project Convening, Invited Talk	2025
NCWIT, Invited Talk	2025
Rising Stars in Data Science Workshop, Talk and Panel	2024 & 2025
Association for the Study of Higher Education, Talk	Nov 2024
University of Colorado Boulder Academic Leadership Conference, Invited Talk	Aug 2024
Athena Fellowship for Girls in STEM Panel, Invited Talk	Aug 2024
International Conference on the Science of Science & Innovation, Talk	July 2024
Kilpatrick Group, University of Colorado Boulder Applied Mathematics, Invited Talk	Apr 2024
Academic Analytics Research Center, Invited Talk	Feb 2024
University of Colorado Boulder Computer Science Research Expo, Poster	Feb 2024
Expo Winner	
University of Colorado Boulder Office of Faculty Affairs, Invited Talk	Dec 2023
International Conference on Computational Social Science, Poster	July 2023
Atlanta Conference on Science & Innovation Policy, Talk	May 2023
Women in Network Science & Diversify NetSci Satellite, Talk	July 2022
International Conference on Computational Social Science, Talk	July 2022
International Conference on the Science of Science & Innovation, Talk	June 2022

Science and innovation policy

Atlanta Conference on Science & Innovation Policy, undergraduate session	2025
Atlanta Conference on Science & Innovation Policy, grants session	2025

Measuring community-level risk of declining healthcare access

Coding it Forward Public Technology Showcase, Talk [12 fellows selected]	Aug 2024

Quantifying inequitable education pathways to scientific and technical careers

Association for the Study of Higher Education, Talk	Nov 2024
International Conference on the Science of Science & Innovation, Talk	July 2024
American Educational Research Association Annual Meeting, Poster	Apr 2024

Quantifying the origins and impacts of book bans in U.S. schools

International Conference on Computational Social Science, Talk	July 2023

The elite undergraduate backgrounds of U.S. professors

International Conference on Computational Social Science, Poster	July 2022
International Conference on the Science of Science & Innovation, Poster	June 2022

Accelerating deep neural networks

International Memory Workshop, Invited Talk	May 2020

Towards detecting dyslexia in children's handwriting using neural networks

American Handwriting Analysis Foundation, Invited Talk — Nov 2019
AI for Social Good Workshop, NeurIPS, Poster — Dec 2019
AI for Social Good Workshop, ICML, Poster — June 2019
Best poster award
AI for Social Good Workshop, ICML, Talk — June 2019
Computer Vision for Global Challenges Workshop, CVPR, Talk — June 2019

Teaching

Guest Lectures, Univ. Colorado — Fall 2024
PHIL 2800: Ethical Puzzles and Moral Conflicts

Professional Development Teaching Assistant, IBM Research — Fall 2019
IBM Research Upskilling Class on Deep Learning
Designed and gave mini-lectures; held office hours

Lead Graduate Assistant Instructor, Indiana Univ. — Fall 2018; Spring 2019
CS C343: Introduction to Data Structures & Algorithms
Designed lab activities, homework assignments, test questions, in-class activities; led lab sections; held office hours and study sessions; evaluated projects, homework and tests; managed a staff of 12 teaching assistants

Undergraduate Teaching Assistant, Indiana Univ. — Spring 2018
CS C343: Introduction to Data Structures & Algorithms
Led lab sections; held office hours and study sessions; evaluated projects, homework and tests

Undergraduate Teaching Assistant, Indiana Univ. — Fall 2016 – Fall 2017
CS C241: Discrete Mathematics for Computer Science
Led lab sections; held office hours and study sessions; evaluated projects, homework and tests

Research Mentoring

Joanna Mendy — 2022 – 2023
Univ. Colorado Sociology & Political Science undergraduate student (2022); Information Science graduate student (2023)
Project: Gendered devaluation underlies faculty retention

Maria Martinez — Summer 2022
Univ. Colorado School of Education graduate student
Project: Gendered devaluation underlies faculty retention

Swag Das — Spring 2022
Univ. Colorado Computer Science undergraduate student
Project: Beliefs about why women leave academia

Jordan Roos — Spring 2022
Univ. Colorado Biomedical Engineering undergraduate student
Project: Academic fields with fewer women are more stressful

Service

Graduate & Family Housing Resident Council Secretary & Treasurer — 2024 – 2025
CU Computer Science PhD Application Mentor — 2020 – 2025
You're @ CU Research Program Graduate Student Mentor — 2022
McNair Scholars Graduate Student Mentor — 2021 – 2022
CU Engineering Mentor for Underrepresented First-Year Undergraduates — 2020 – 2022
Lead Ambassador, IU Luddy School of Informatics, Computing & Engineering — 2016 – 2019

Software Development Intern, Serve IT Nonprofit Technology Clinic 2016 – 2017

Reviewer for: Behaviour Research and Therapy, eLife, The Lancet Regional Health – Americas, Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, Research Policy, Social Policy & Administration

Additional Employment

Research Assistant Sep 2017 – June 2019
Indiana University Computer Vision Lab (Bloomington, IN)

Research Intern Summer 2018
IBM Research, *Artificial Intelligence Hardware Group* (San Jose, CA)

Research Intern Summer 2017
MIT Lincoln Laboratory, *Machine Learning Group* (Boston, MA)

Research Assistant Aug 2016 – Sep 2017
Indiana University Kelley School of Business (Bloomington, IN)

Research Assistant Summer 2016
NSF Research Experience for Undergraduates (Bloomington, IN)

Professional Activities

Atlanta Academy on Science and Innovation Policy at Georgia Tech 2024
Selected participant. Funded by Georgia Tech School of Public Policy.

Summer Institute in Computational Social Science at Duke University 2022
Selected participant. Funded by the Russell Sage Foundation.

Grad Cohort for Women Workshop, Computing Research Association (CRA) 2021
Selected participant. (Remote)